

AASRA — Complete Technology Stack Specification

Detailed Technology Breakdown: Google Cloud Platform vs Other Component Tech

Target Platform: Syngenta Biologicals Overwatch & AASRA Agricultural Intelligence

Architectural Audit Date: August 14, 2026 | **Scope:** 100% Codebase Component Audit (Frontend, Backend, ML, Weather, GIS)

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This document presents the complete technical stack powering **AASRA** (Agricultural Intelligence & Yield Protection Platform). AASRA utilizes **Google Cloud Platform (GCP)** and **Google AI** as its core intelligence, voice, multimodal vision, and serverless compute layer. In addition to Google Cloud, AASRA leverages a curated set of specialized open-source frameworks, geospatial engines, weather telemetry APIs, and relational databases. Below is the exhaustive, categorized specification of all technology used across the platform.

1. Primary Google Cloud & Google AI Technology Stack

AASRA is engineered around a **100% Google Cloud AI foundation** for multilingual conversational intelligence, vision diagnostics, high-definition speech synthesis, and serverless deployment.

- **Google Cloud Run [Serverless Container Execution]:** Hosts the AASRA FastAPI Gateway (Python 3.11 / Uvicorn container). Provides auto-scaling, low-latency HTTPS REST routing, zero cold starts, and container isolation.
- **Google Gemini 2.0 Flash (gemini-2.0-flash) [Multilingual LLM & RAG Reasoning Engine]:** Powers Problem Statement 04 (PS-04). Provides multi-turn RAG advisories across 12 Indian languages (Hindi, Marathi, Telugu, Tamil, Kannada, Malayalam, Gujarati, Punjabi, Bengali, Odia, Assamese, English) with explainable rationales and confidence scores (92-96%).
- **Google Gemini 2.0 Flash Vision [Multimodal Crop Leaf Diagnostics]:** Scans plant foliage photos uploaded by farmers (`/api/chat/diagnose-leaf`). Detects leaf necrosis, chlorosis, thermal scorch, and foliar diseases while synthesizing prescriptive Syngenta biostimulant treatment dosages.
- **Google Chirp 3 HD Speech API [Neural Text-to-Speech (TTS)]:** Generates high-definition regional accent audio streams (`/api/chat/google-tts`) delivering vocalized advisories to farmers directly in their native language.
- **Google Speech-to-Text (STT) & Speech API [Voice Query Ingestion]:** Handles real-time acoustic voice input from farmers in the `/ask` voice assistant interface, converting spoken regional dialects into structured prompt context.
- **Google Cloud REST Generative Language API [Transport & Multi-Key Routing]:** Underlying REST protocol (`google-generativeai` SDK v0.8.3). Includes backend multi-key auto-rotation strategy across 4 API keys (`GOOGLE\_API\_KEY\_1..4`) for rate-limit and quota resilience.

2. Other / Additional Technology Stack Used in AASRA

To deliver a complete end-to-end enterprise web application, AASRA integrates specialized frontend, backend, database, geospatial, weather, and machine learning technologies alongside Google Cloud:

2.1 Frontend Framework & User Interface Tier

- **Next.js 16.3.0 (React 19.2.8) [Full-Stack Web Framework]**: App Router architecture (`frontend/src/app`) providing server-side rendering, client hydration, and page routing (`/`, `/ask`, `/fields`, `/journal`, `/impact`, `/what-if`).
- **TypeScript 5 [Static Type Safety]**: Strict type contracts across React components, API request/response payloads, and state stores.
- **Tailwind CSS v4 (@tailwindcss/postcss) [Design System & Styling]**: Vanilla CSS-first utility architecture implementing a dark-mode Linear-inspired visual style (`design-md-linear.app`) with glassmorphic cards and emerald accents.
- **Framer Motion 13.1.0 & Lucide React [Animations & Icons]**: Micro-animations for stress gauges, card expansions, modal transitions, and SVG icon system.
- **Recharts 3.10.1 [Data Visualization]**: Renders interactive yield attribution charts, Return on Biological Investment (ROBI) trend lines, and weather historical plots.
- **Web Speech API [Browser Audio Fallback]**: Provides client-side fallback text-to-speech synthesis (`SpeechSynthesisUtterance`) with BCP-47 regional language tags when offline.

2.2 Geospatial GIS & Field Mapping Engine

- **Leaflet.js 1.9.4 & Leaflet Draw [Interactive GIS Map Editor]**: Powers the `/fields` field polygon manager. Enables farmers to draw custom field boundaries, calculate acreage using the Shoelace area algorithm, and render interactive heatmap overlays (Temperature, Drought, Crop Health).

2.3 Backend Application Tier & Async Microservices

- **Python 3.11 [Runtime Environment]**: Core backend execution environment for mathematical algorithms, RAG pipelines, and data orchestration.
- **FastAPI 0.115.0 & Uvicorn 0.30.6 [ASGI Web Server & REST Gateway]**: Asynchronous high-performance REST backend handling incoming client traffic (`backend/app/main.py`).
- **Pydantic 2.9.2 & Pydantic-Settings [Data Validation & Config]**: Strict schema validation for API requests/responses and environment secret loading (`backend/app/config.py`).
- **HTTPX 0.27.2 [Async HTTP Client]**: Handles asynchronous outbound requests to external weather and agronomic APIs with non-blocking concurrency.
- **SQLAlchemy 2.0.35 & Alembic 1.13.3 [Database ORM & Migrations]**: Relational database object mapping and schema migration management (`backend/app/database.py`).
- **Python-Jose & Passlib (bcrypt) [Security & Authentication]**: JWT token encoding/decryption and cryptographic hashing for secure user access control.
- **NumPy 1.26.4 & SciPy 1.13.1 [Scientific Computing & Math]**: Executes biophysical differential equations, degree-hour aggregations, and crop yield degradation curve modelling.

2.4 External Data Telemetry & Agronomic APIs

- **Meteoblue Dataset API [Meteorological & Climate Telemetry]**: Provides hyper-local weather datasets via `NEMSGLOBAL` (10-day forecast) and `ERA5` (historical reanalysis 1940–present). Fetches 2m temperature (Code 11), precipitation (Code 61), soil moisture 0-10cm (Code 144), and evapotranspiration (Code 261).
- **Syngenta CE Hub API [Syngenta Decision Engine]**: Provides enterprise agronomic recommendations: Growing Degree Days (GDD), Hydric Stress, Biological & Chemical Spray Windows, Planting Windows, Chilling Units, and

Disease Risk metadata.

2.5 Databases, Data Persistence & Machine Learning

- **PostgreSQL Database (psycpg2-binary 2.9.9) [Relational Database]**: Persistent relational storage for registered user profiles, field records, and biological intervention logs.
- **JSON Field Database (fields_db.json) [Spatial Polygon Storage]**: Lightweight file-based persistent cache storing GeoJSON polygon spatial coordinates and calculated field acreage.
- **5-Layer Hybrid ML Stress Engine (ps02-engine) [Scikit-Learn ML Stress Model]**: Python-based Scikit-Learn (Random Forest, SVM) stress classification engine, Flask/Streamlit prototype, and 9 deterministic agronomic stress/ROBI algorithms.

3. Summary Technology Matrix & Stack Classification

The following matrix summarizes every technology used in AASRA, explicitly categorizing whether it belongs to the **Google Cloud / Google AI** ecosystem or the **Additional System Stack**.

Category	Technology Name	Ecosystem	Role & Function in AASRA
Cloud Compute	Google Cloud Run	Google Cloud	Serverless container execution & FastAPI backend hosting
Generative AI	Gemini 2.0 Flash	Google Cloud	Multilingual 12-language RAG advisory & reasoning engine
Vision AI	Gemini 2.0 Flash Vision	Google Cloud	Multimodal foliar disease scan & biostimulant prescription
Speech Synthesis	Google Chirp 3 HD	Google Cloud	High-definition neural regional Text-to-Speech (TTS)
Voice Ingestion	Google Speech-to-Text	Google Cloud	Real-time farmer audio voice processing in /ask
AI SDK & API	Google GenAI SDK (0.8.3)	Google Cloud	Python SDK & REST transport with multi-key auto-rotation
Frontend Framework	Next.js 16 / React 19	Open Source	App Router full-stack web application framework
Frontend Language	TypeScript 5	Microsoft / Open	Static typing across components and backend schemas
UI Styling	Tailwind CSS v4	Open Source	Linear-inspired dark-mode styling & glassmorphic UI
Geospatial GIS	Leaflet.js & Leaflet Draw	Open Source	Interactive field polygon drawing, area & heatmaps
Backend Server	FastAPI / Uvicorn	Open Source	Python 3.11 async REST API gateway and endpoints
Database ORM	SQLAlchemy / Alembic	Open Source	Relational database object mapping and migrations
Relational DB	PostgreSQL (psycpg2)	Open Source	Persistent relational store for fields & intervention logs
Scientific Math	NumPy & SciPy	Open Source	Agronomic yield degradation curve & stress algorithms
Weather Data	Meteoblue Dataset API	Meteoblue AG	Live weather telemetry & ERA5 climate reanalysis
Agronometrics API	Syngenta CE Hub API	Syngenta	GDD, Hydric Stress, Spray & Planting window recommendations
Hybrid ML Model	Scikit-Learn (RF/SVM)	Open Source	5-layer ML climate stress engine prototype (ps02-engine)

4. Architectural Compliance & Technology Summary

Summary Verdict: AASRA strictly complies with the requirement for a **100% Google Cloud AI Core**. All artificial intelligence, multilingual natural language understanding (NLU), multimodal foliar computer vision, and neural speech synthesis operations are powered exclusively by Google Cloud endpoints (Gemini 2.0 Flash, Chirp 3 HD, STT).

To support these Google Cloud AI capabilities in production, AASRA incorporates robust supporting technologies: Next.js 16/React 19 for the frontend, FastAPI for the backend gateway, Leaflet.js for GIS field mapping, PostgreSQL for relational persistence, and Meteoblue / Syngenta CE Hub APIs for real-time agronomic data ingestion.